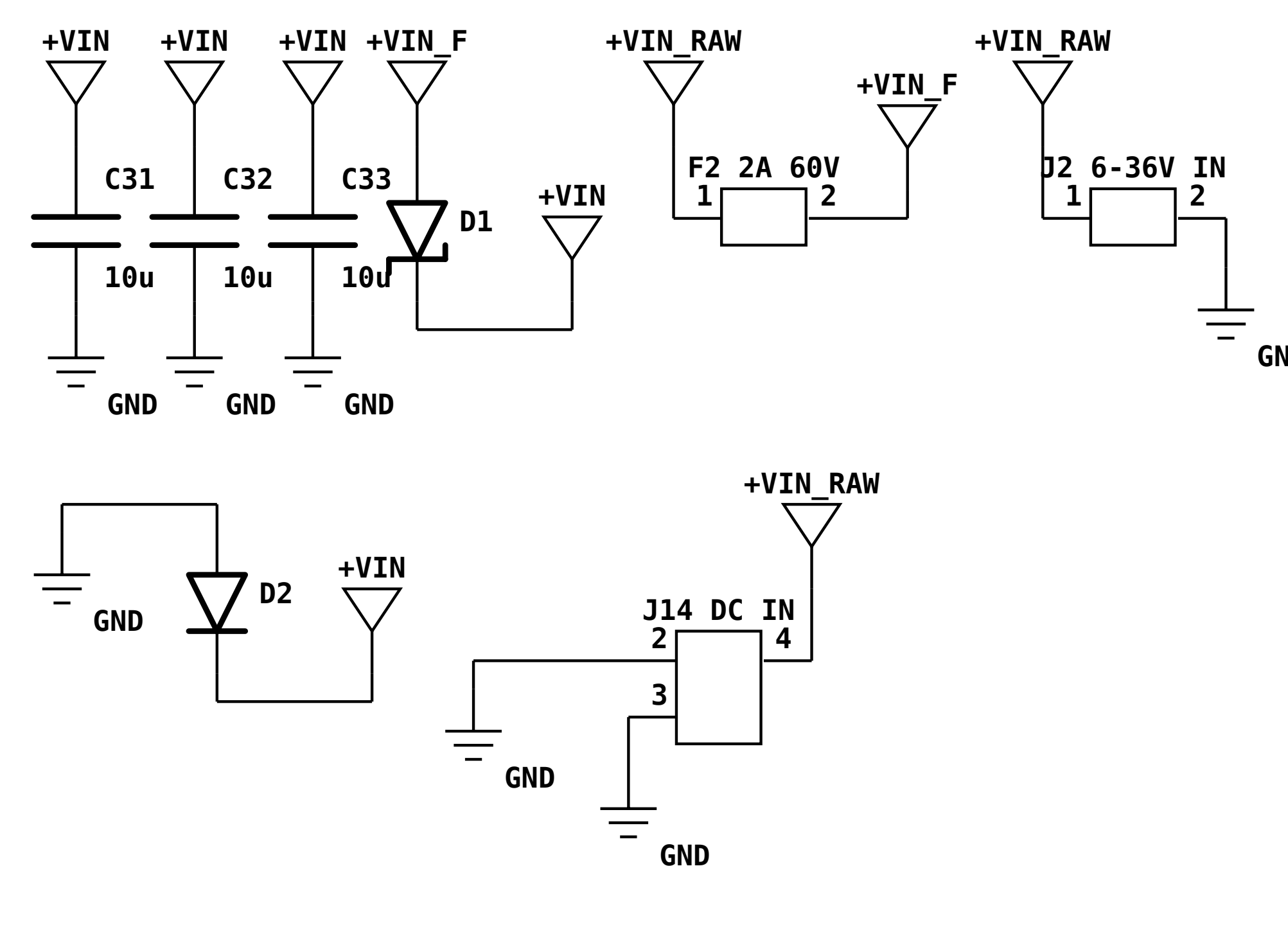
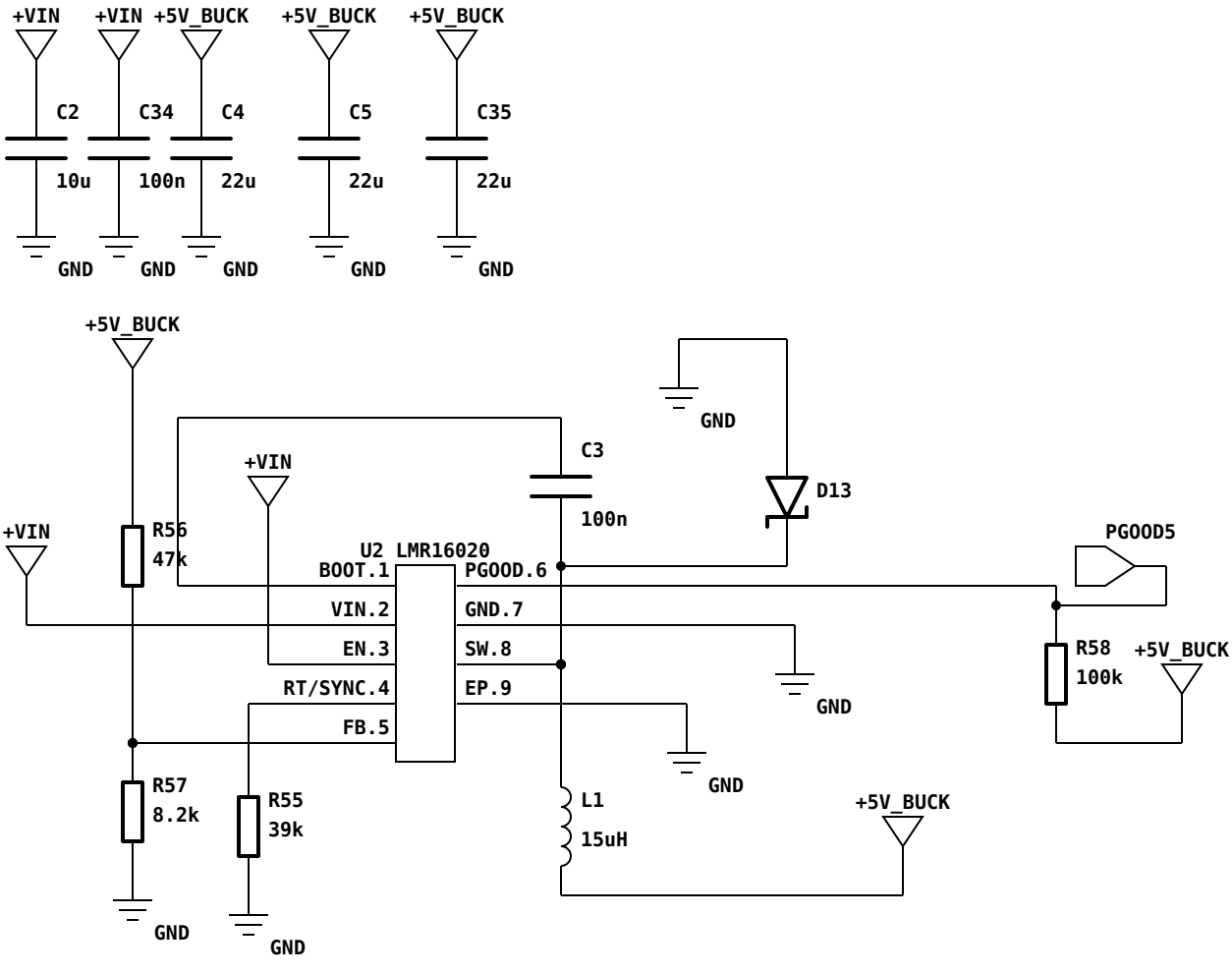
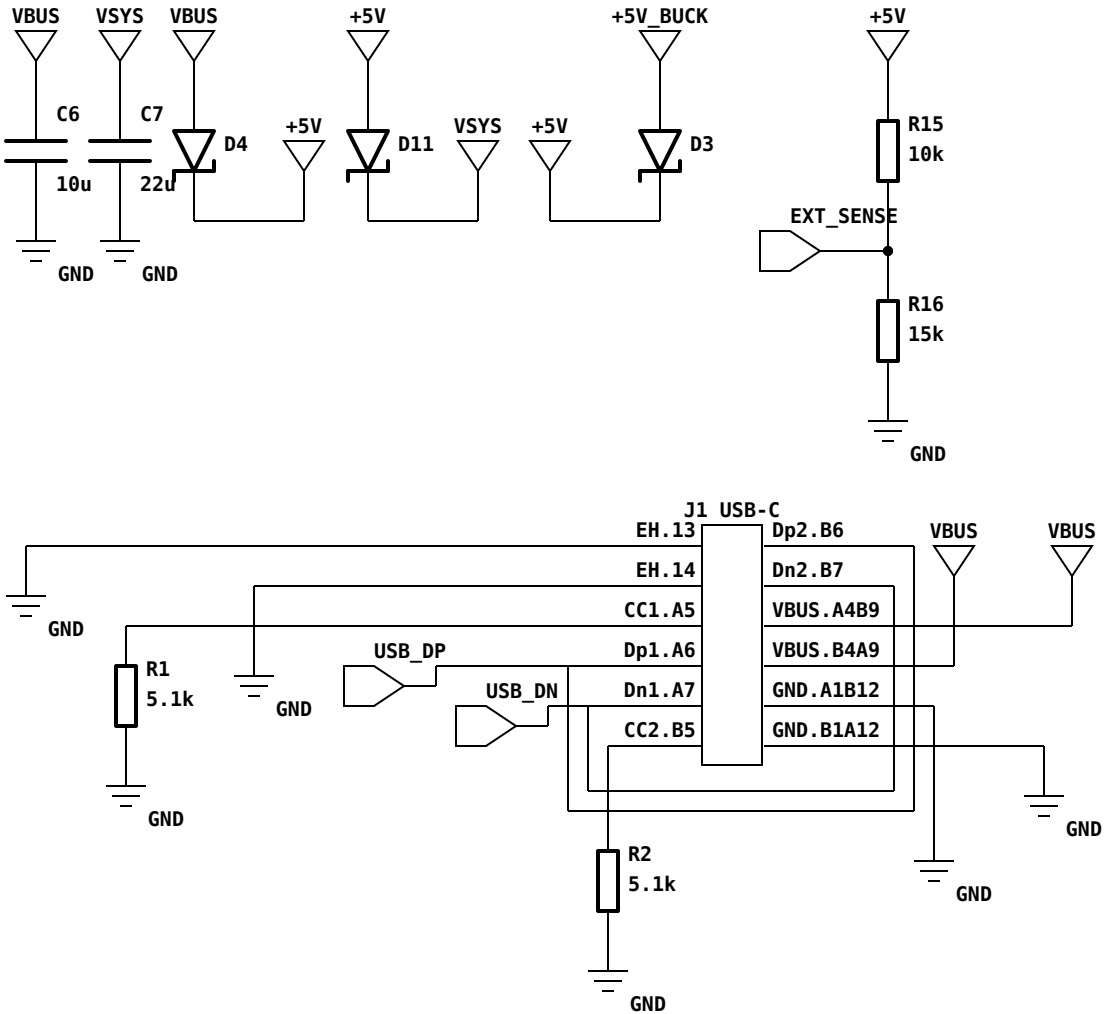
Wiegand reader bridge — revision 10 schematic

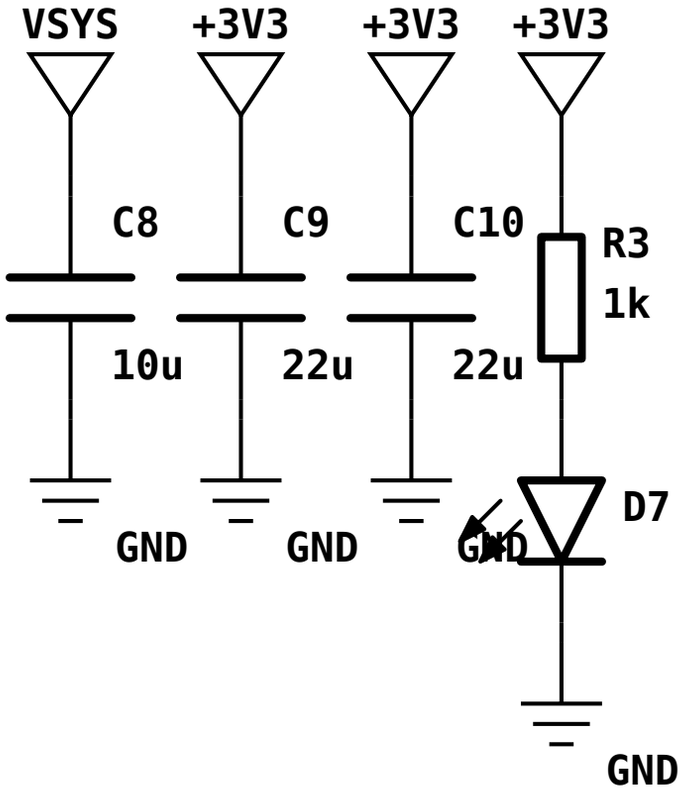
Generated from netlist.py (netlistsvg). Supply nets are flags; nets shared between blocks are named ports. I

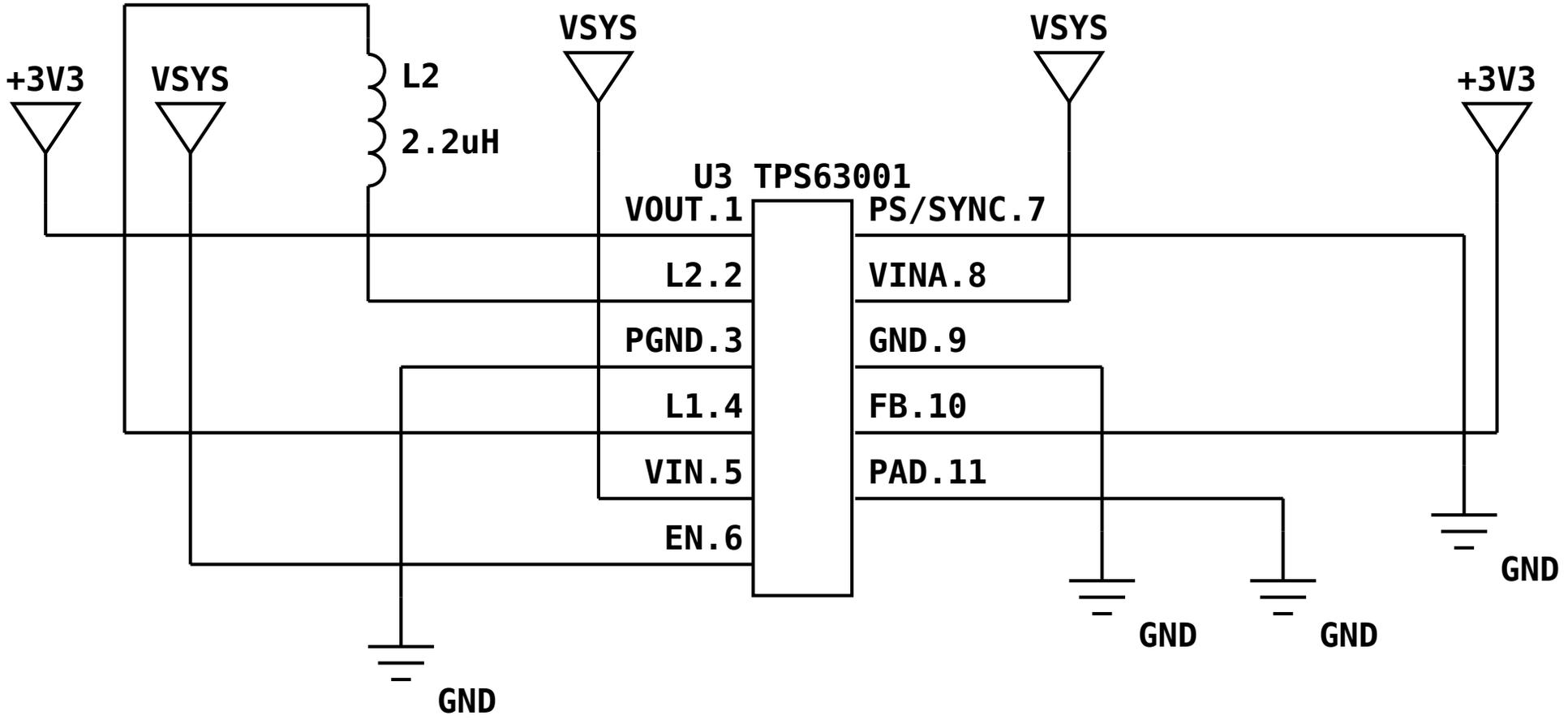
1. Block 1: 6-36 V input
2. Block 2: buck to 5 V (LMR16020, 60 V; datasheet SNVSAH8A sec. 8.2)
3. Block 3: USB-C, +5V, VSYS
4. Block 4: 3.3 V buck-boost
5. Block 5: MCU
6. Block 6: Wiegand inputs
7. Block 7: LED and BEEP
8. Block 8: reader supply
9. Block 9: battery
10. Block 10: screens
11. Block 11: inputs
12. Block 12 (rev 10): I/O expander, TCA9535 at 0x20
13. Block 13 (rev 10): relay -- driver fitted, relay and terminals NOT fitted
14. Block 14 (rev 10): RS-485 / OSDP on UART0
15. Block 15 (rev 10): surge/ESD on every corridor wire (SM712: +12 / -7 V standoff)
16. Block 16 (rev 10): reader current sense
17. Block 17 (rev 10): W5500 module header and Qwiic, not fitted

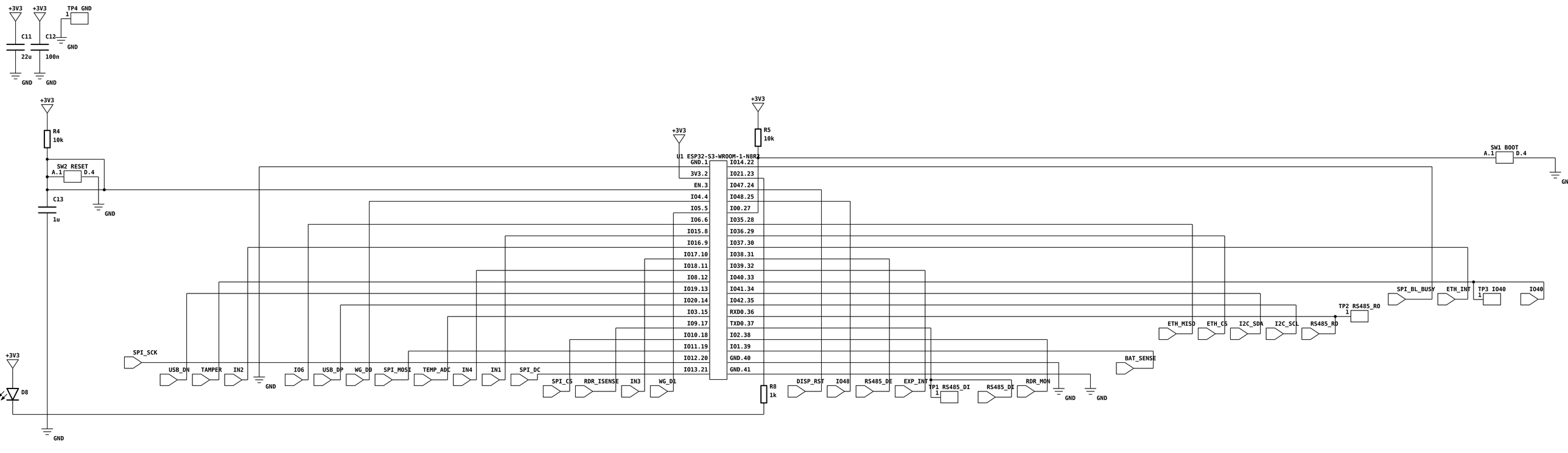

+VIN
+VIN
+VIN
+VIN_F
C31
C32
C33
10u
10u
10u
D1
+VIN
GND
GND
GND
+VIN_RAW
+VIN_F
F2 2A 60V
1
2
+VIN_RAW
J2 6-36V IN
1
2
GN
GND
D2
+VIN
+VIN_RAW
J14 DC IN
2
4
3
GND
GND

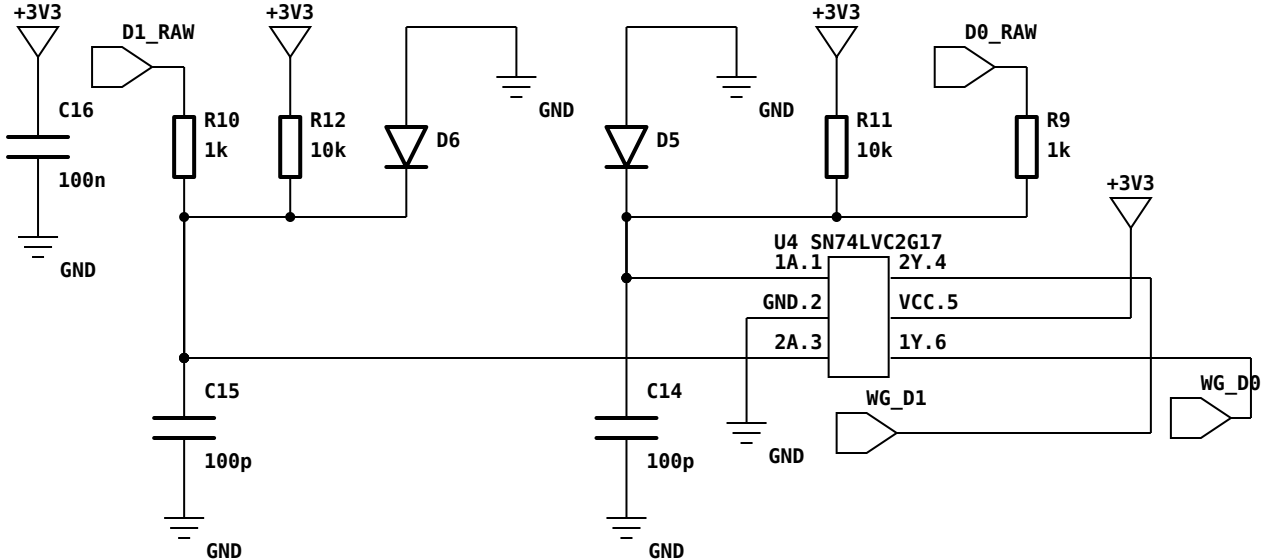

+VIN
+VIN
+5V_BUCK
+5V_BUCK
+5V_BUCK
C2
C34
C4
C5
C35
10u
100n
22u
22u
22u
GND
GND
GND
GND
GND
+5V_BUCK
+VIN
+VIN
R56
47k
C3
100n
GND
D13
U2 LMR16020
BOOT.1
VIN.2
EN.3
RT/SYNC.4
FB.5
PGOOD.6
GND.7
SW.8
EP.9
PGOOD5
R58
100k
+5V_BUCK
GND
GND
R57
8.2k
R55
39k
L1
15uH
+5V_BUCK
GND
GND

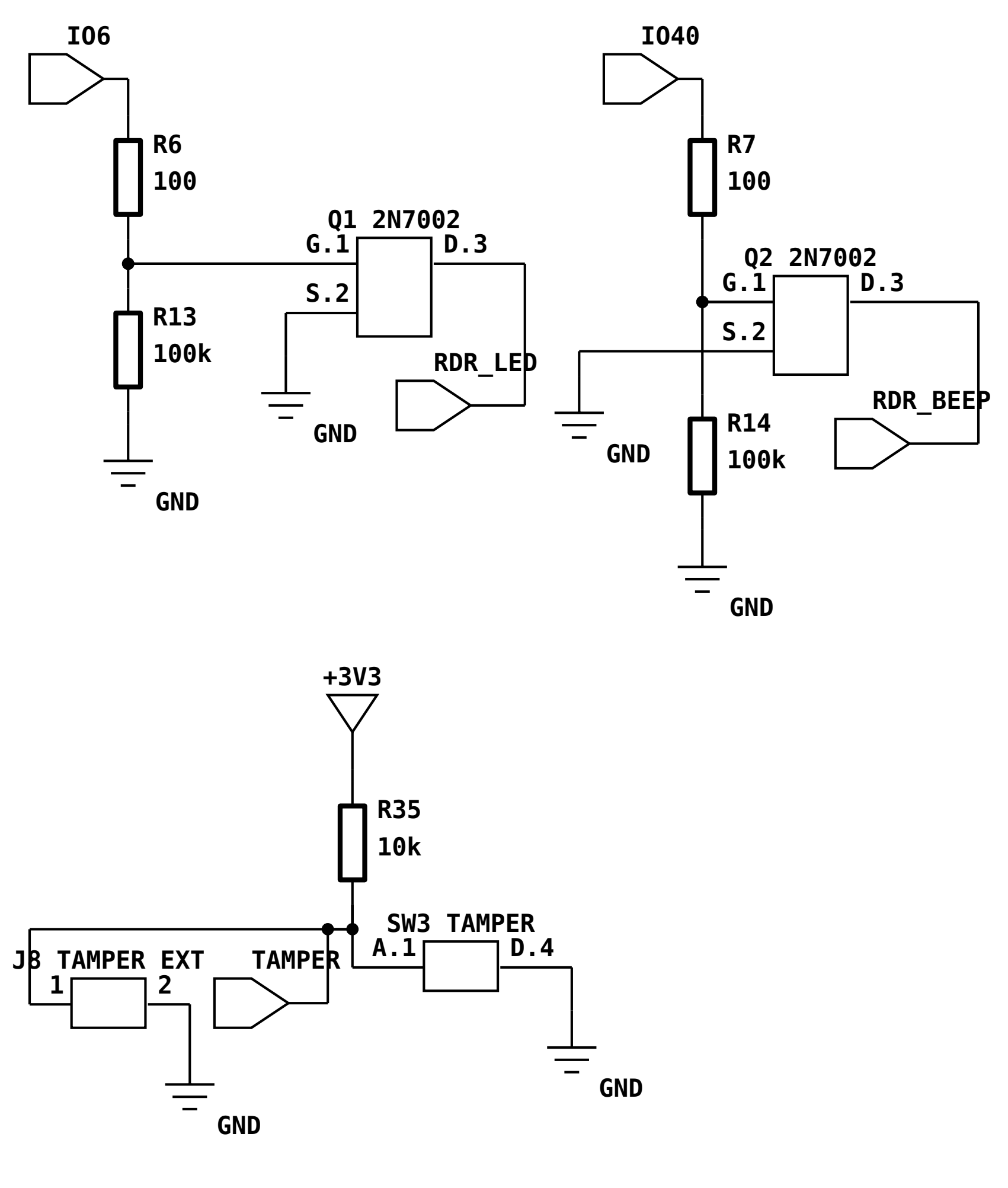

VBUS
VSYS
VBUS
+5V
+5V_BUCK
+5V
C6
10u
C7
22u
D4
+5V
D11
VSYS
+5V
D3
R15
10k
EXT_SENSE
R16
15k
GND
GND
GND
GND
J1 USB-C
EH.13
EH.14
CC1.A5
Dp1.A6
Dn1.A7
CC2.B5
Dp2.B6
Dn2.B7
VBUS.A4B9
VBUS.B4A9
GND.A1B12
GND.B1A12
VBUS
VBUS
GND
R1
5.1k
GND
USB_DP
USB_DN
GND
GND
GND
R2
5.1k
GND

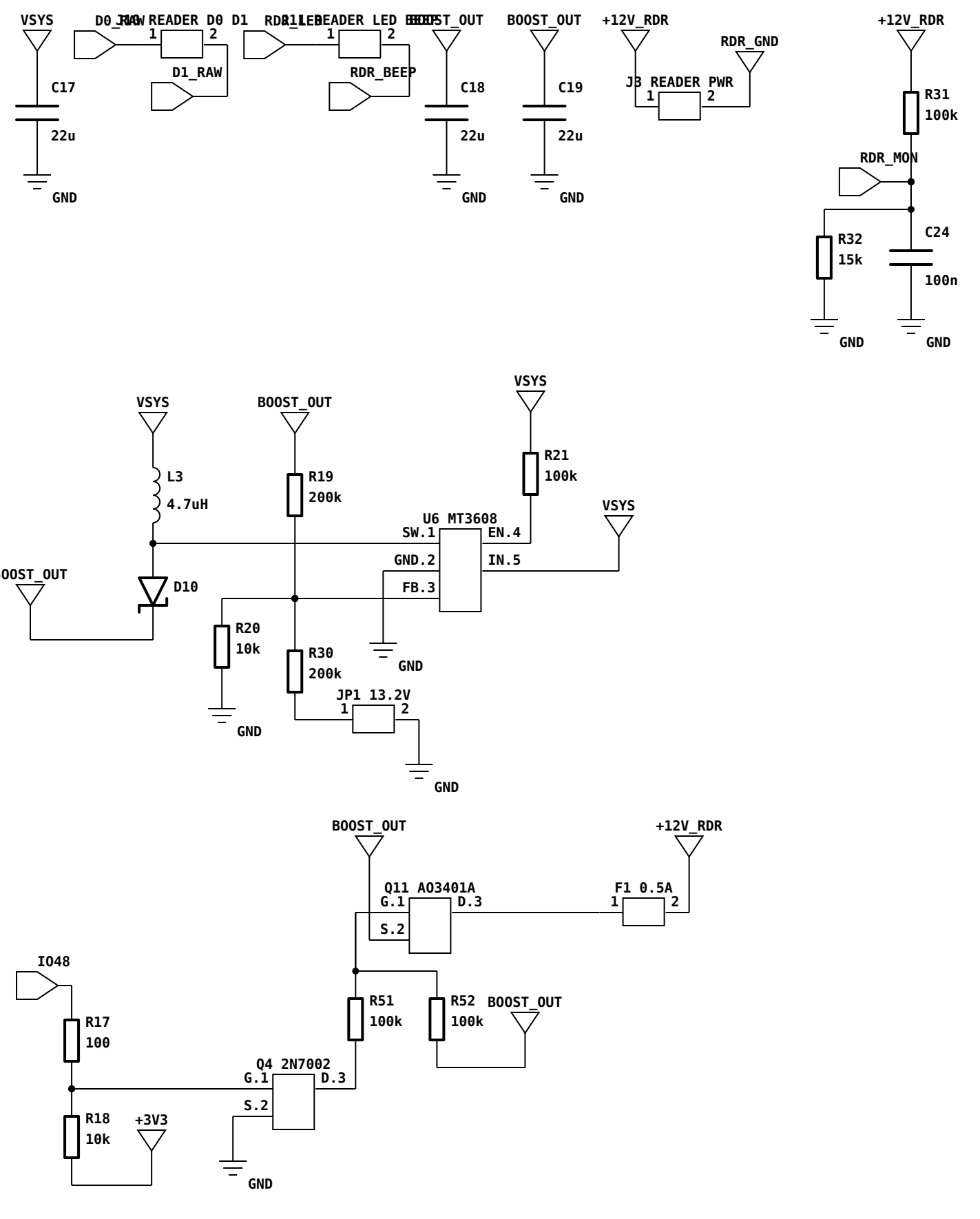

VSYS
+3V3
+3V3
+3V3
C8
C9
C10
R3
1k
10u
22u
22u
GND
GND
GND
D7
GND

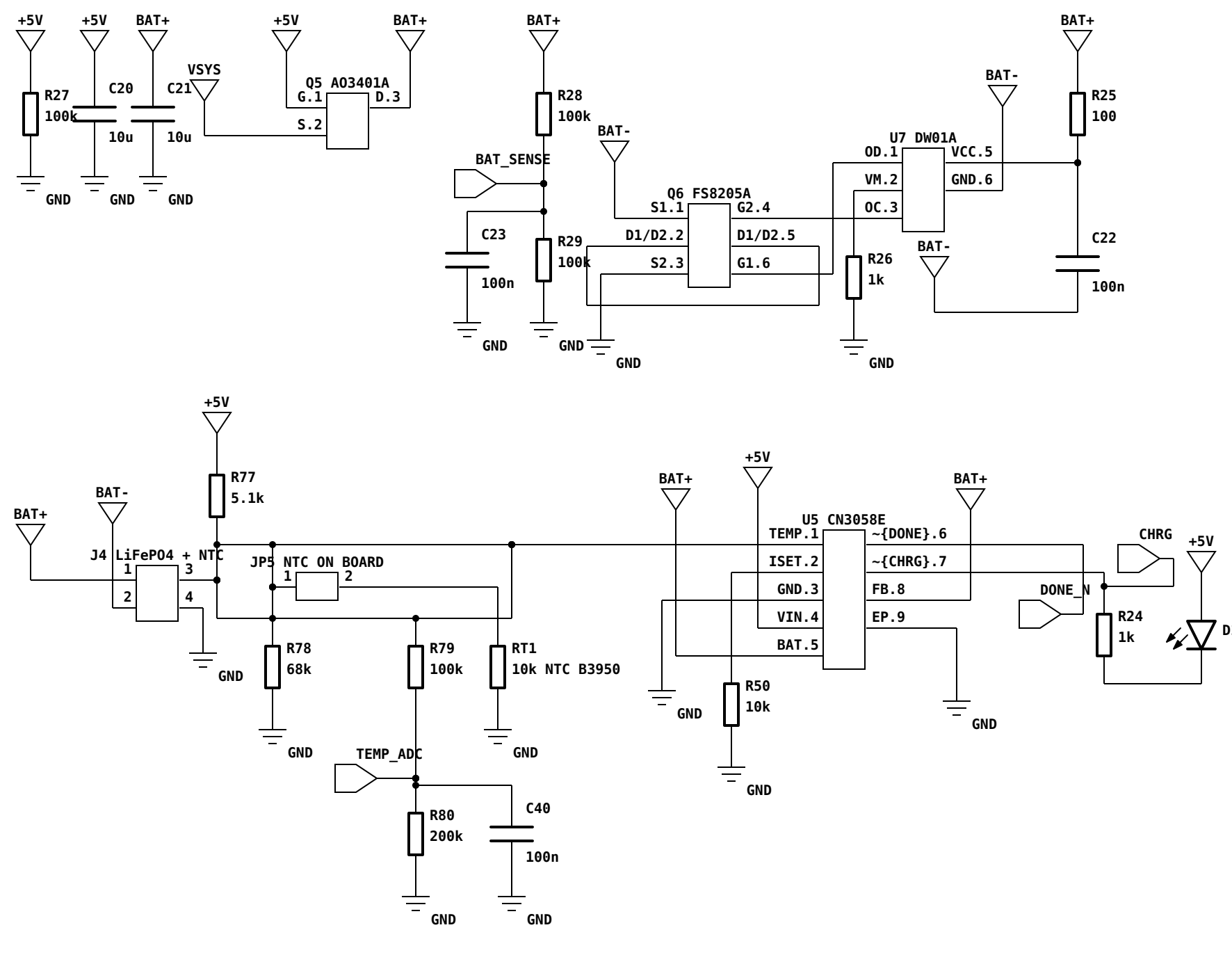

+3V3
VSYS
L2
2.2uH
VSYS
VSYS
+3V3
U3 TPS63001
VOUT.1
PS/SYNC.7
L2.2
VINA.8
PGND.3
GND.9
L1.4
FB.10
VIN.5
PAD.11
EN.6
GND
GND
GND
GND

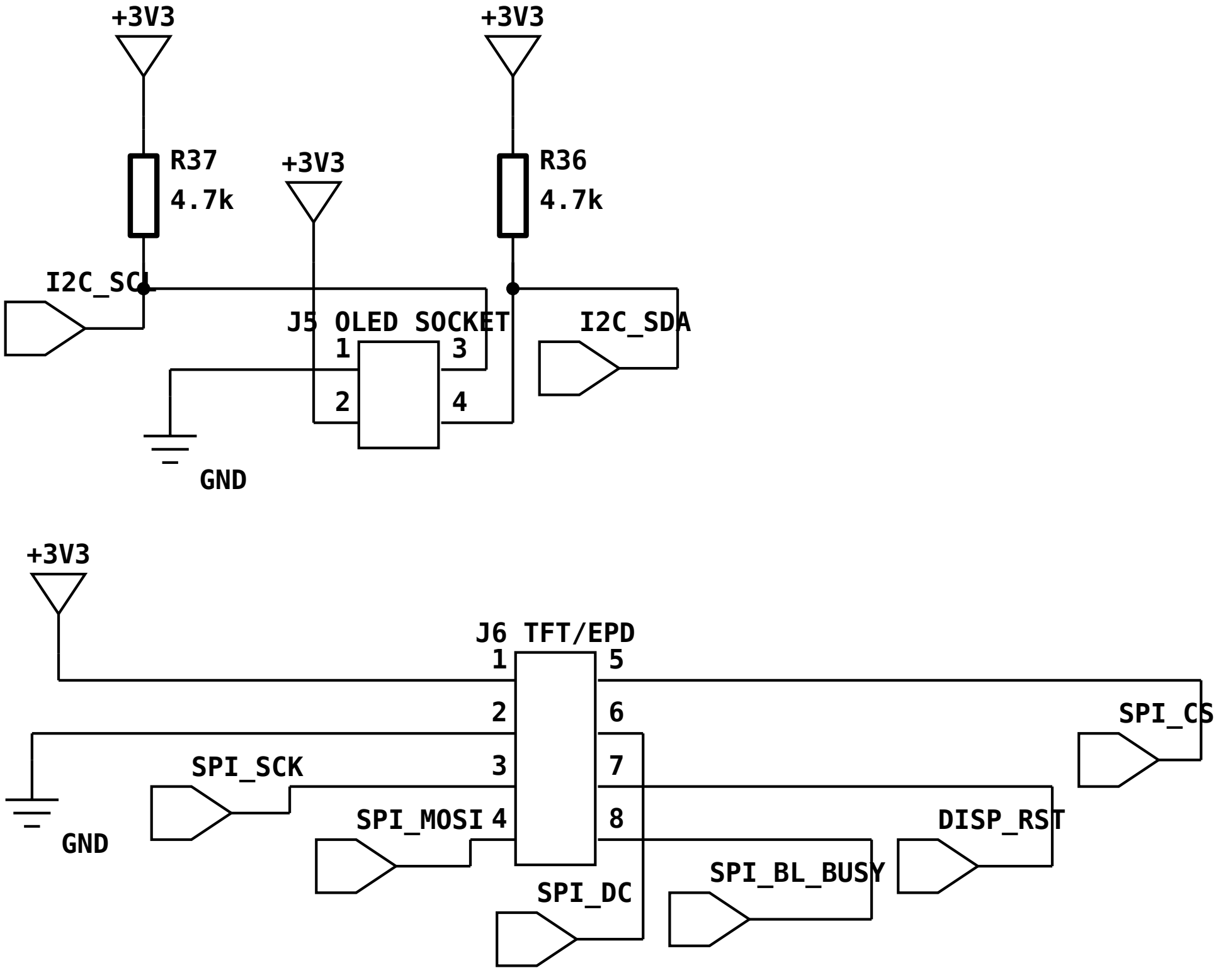

+3V3
+3V3
TP4 GND
1
C11
C12
GND
22u
100n
GND
GND
+3V3
R4
10k
SW2 RESET
A.1
D.4
C13
1u
GND
GND
+3V3
D8
+3V3
R5
10k
U1 ESP32-S3-WROOM-1-N8R2
GND.1
3V3.2
EN.3
IO4.4
IO5.5
IO6.6
IO15.8
IO16.9
IO17.10
IO18.11
IO8.12
IO19.13
IO20.14
IO3.15
IO9.17
IO10.18
IO11.19
IO12.20
IO13.21
IO14.22
IO21.23
IO47.24
IO48.25
IO0.27
IO35.28
IO36.29
IO37.30
IO38.31
IO39.32
IO40.33
IO41.34
IO42.35
RXD0.36
TXD0.37
IO2.38
IO1.39
GND.40
GND.41
R8
1k
SW1 BOOT
A.1
D.4
GND
SPI_SCK
USB_DN
TAMPER
IN2
GND
IO6
USB_DP
WG_D0
SPI_MOSI
TEMP_ADC
IN4
IN1
SPI_DC
SPI_CS
RDR_ISENSE
IN3
WG_D1
DISP_RST
IO48
RS485_DE
EXP_INT
TP1 RS485_DI
1
RS485_DI
RDR_MON
GND
GND
BAT_SENSE
ETH_MISO
ETH_CS
I2C_SDA
I2C_SCL
RS485_RO
TP2 RS485_RO
1
RS485_RO
SPI_BL_BUSY
ETH_INT
TP3 IO40
1
IO40

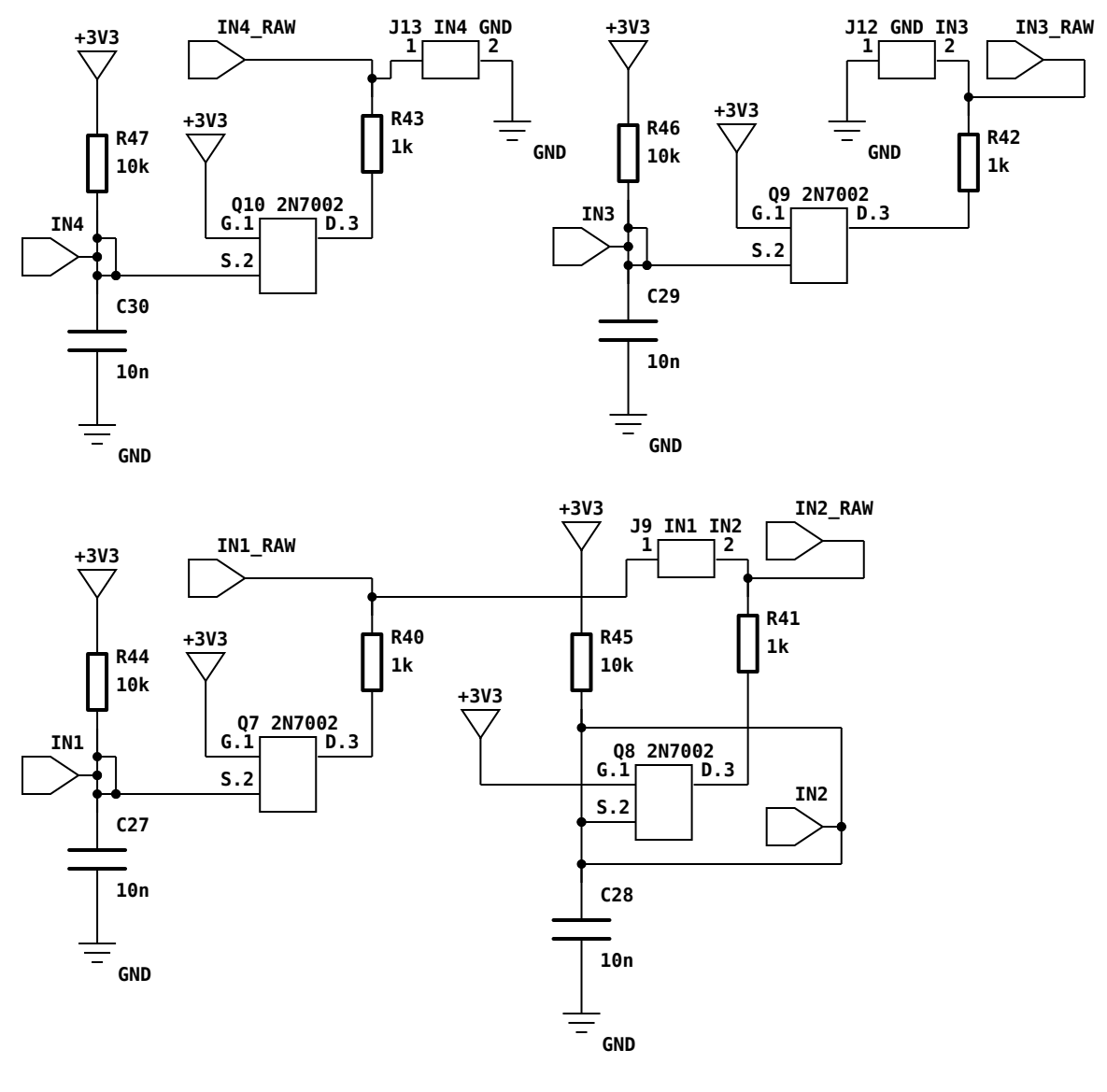

+3V3
C16
100n
GND
D1_RAW
R10
1k
+3V3
R12
10k
D6
GND
D5
GND
+3V3
R11
10k
D0_RAW
R9
1k
+3V3
U4 SN74LVC2G17
1A.1
2Y.4
GND.2
VCC.5
2A.3
1Y.6
C15
100p
GND
C14
100p
GND
GND
WG_D1
WG_D0

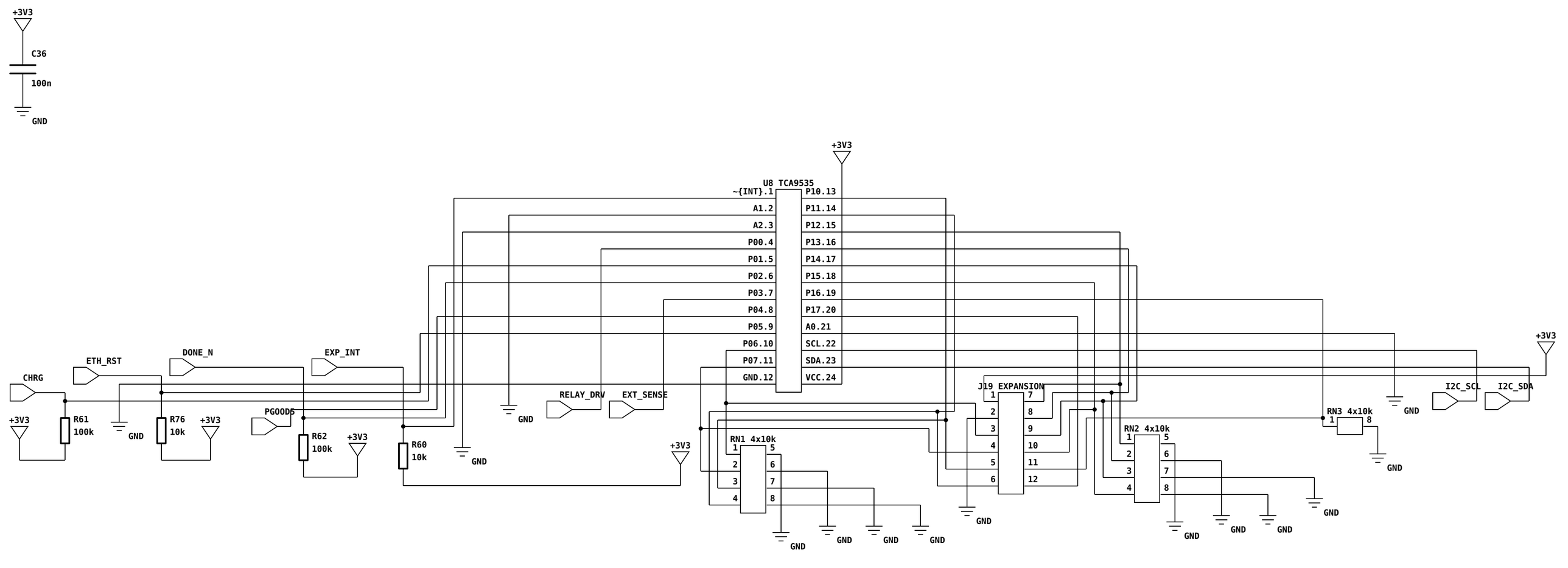

IO6
R6
100
Q1 2N7002
G.1
D.3
S.2
R13
100k
RDR_LED
GND
GND
IO40
R7
100
Q2 2N7002
G.1
D.3
S.2
RDR_BEEP
R14
100k
GND
GND
+3V3
R35
10k
SW3 TAMPER
A.1
D.4
J8 TAMPER EXT
1
2
TAMPER
GND
GND

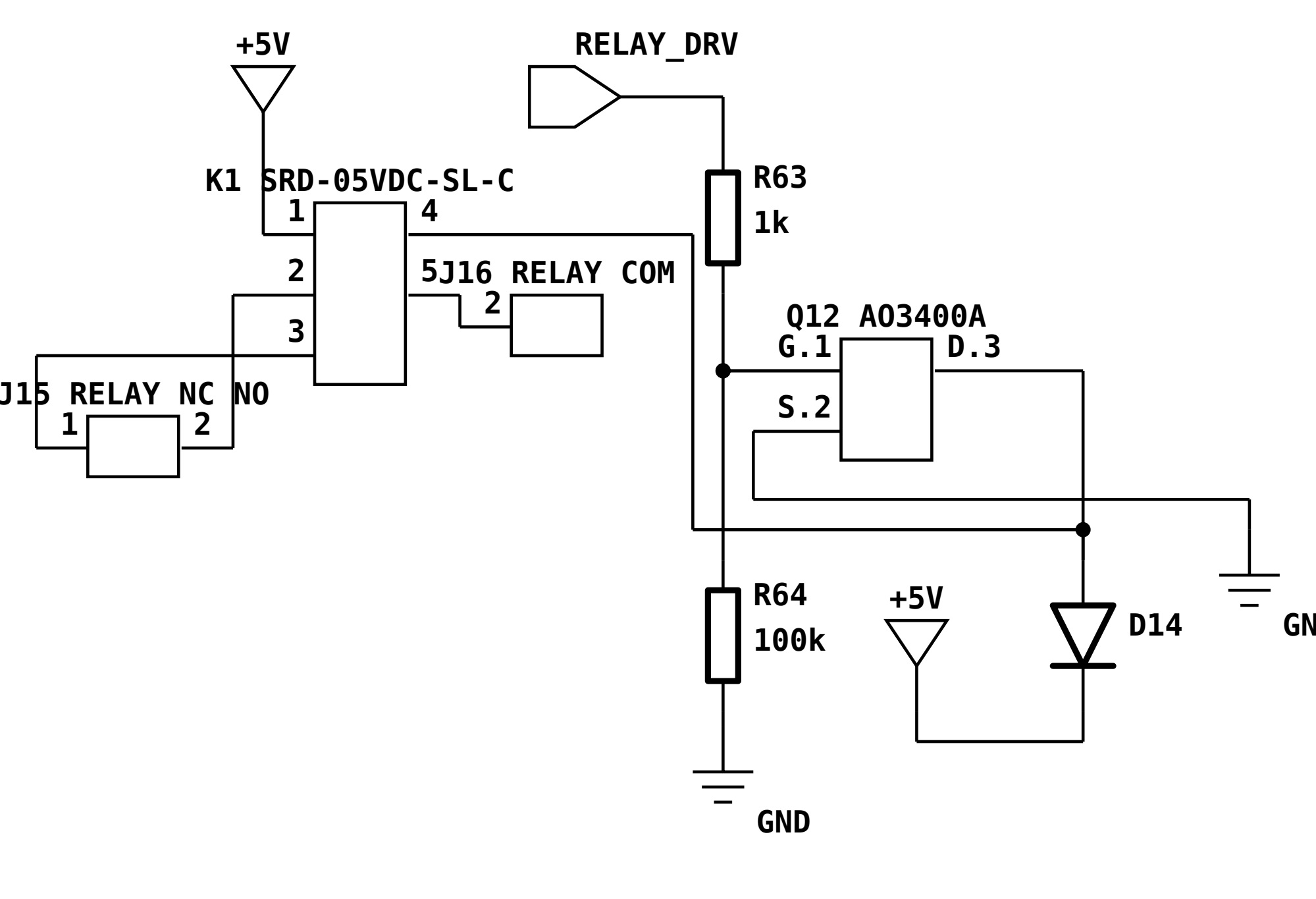

VSYS
C17
22u
GND
D0_RAW
J10 READER D0 D1
1
2
D1_RAW
RDR_LED
J11 READER LED BEEP
1
2
RDR_BEEP
BOOST_OUT
C18
22u
GND
BOOST_OUT
C19
22u
GND
+12V_RDR
J8 READER PWR
1
2
RDR_GND
+12V_RDR
R31
100k
RDR_MON
R32
15k
C24
100n
GND
GND
VSYS
L3
4.7uH
BOOST_OUT
R19
200k
VSYS
R21
100k
U6 MT3608
SW.1
EN.4
GND.2
IN.5
FB.3
VSYS
BOOST_OUT
D10
R20
10k
R30
200k
GND
JP1 13.2V
1
2
GND
GND
BOOST_OUT
+12V_RDR
Q11 AO3401A
G.1
D.3
S.2
F1 0.5A
1
2
IO48
R17
100
R51
100k
R52
100k
BOOST_OUT
Q4 2N7002
G.1
D.3
S.2
R18
10k
+3V3
GND

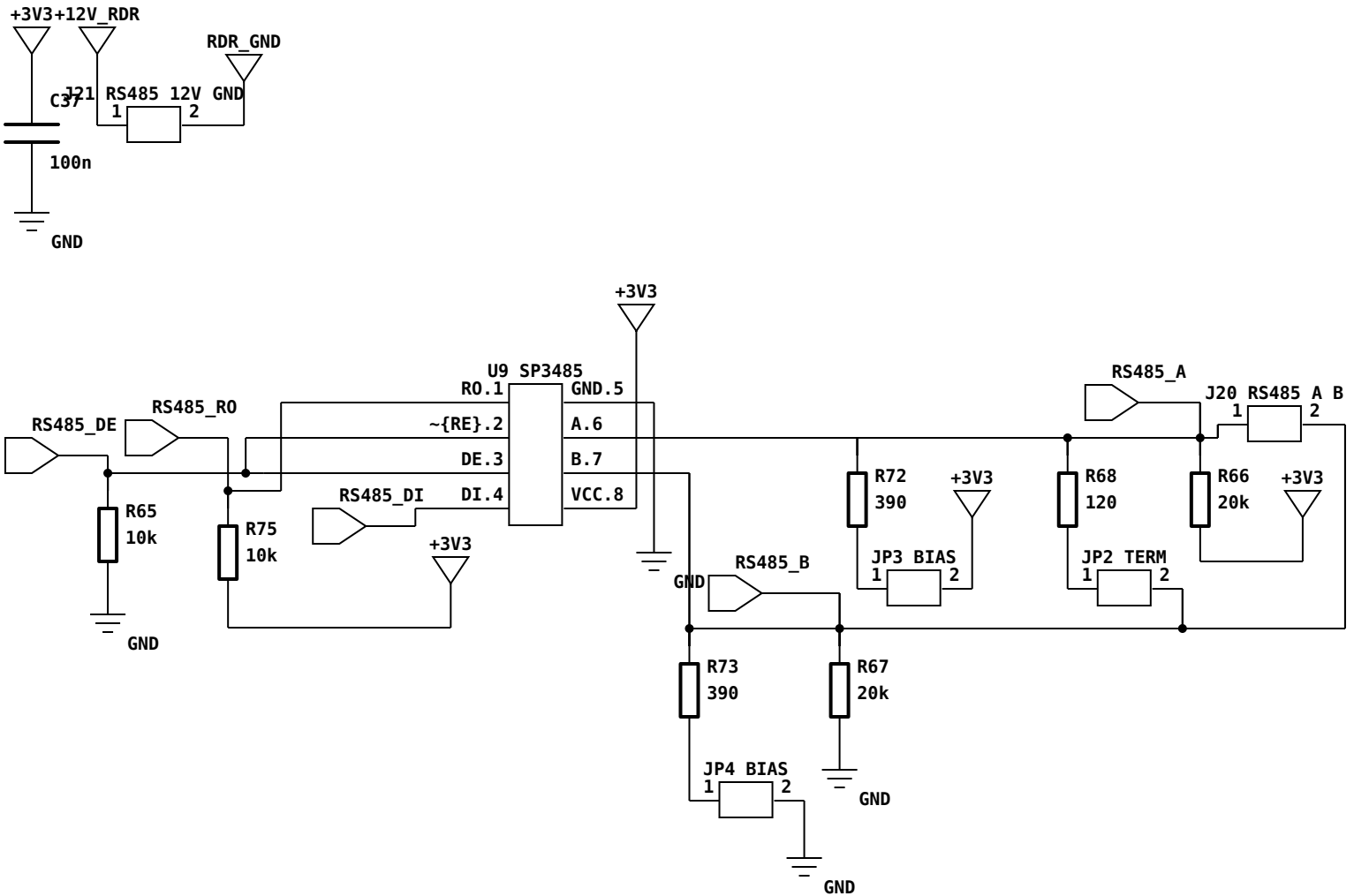

+5V
+5V
BAT+
+5V
BAT+
R27
100k
C20
10u
C21
10u
VSYS
Q5 AO3401A
G.1
D.3
S.2
GND
GND
GND
BAT+
BAT-
BAT+
R28
100k
BAT_SENSE
C23
100n
R29
100k
Q6 FS8205A
S1.1
G2.4
D1/D2.2
D1/D2.5
S2.3
G1.6
GND
GND
GND
BAT-
U7 DW01A
OD.1
VCC.5
VM.2
GND.6
OC.3
R26
1k
GND
BAT+
R25
100
BAT-
C22
100n
+5V
R77
5.1k
BAT-
BAT+
J4 LiFePO4 + NTC
1
2
3
4
JP5 NTC ON BOARD
1
2
R78
68k
R79
100k
RT1
10k NTC B3950
GND
GND
GND
TEMP_ADC
R80
200k
C40
100n
GND
GND
+5V
BAT+
BAT+
U5 CN3058E
TEMP.1
ISET.2
GND.3
VIN.4
BAT.5
~{DONE}.6
~{CHRG}.7
FB.8
EP.9
R50
10k
GND
GND
CHRG
+5V
DONE_N
R24
1k
D

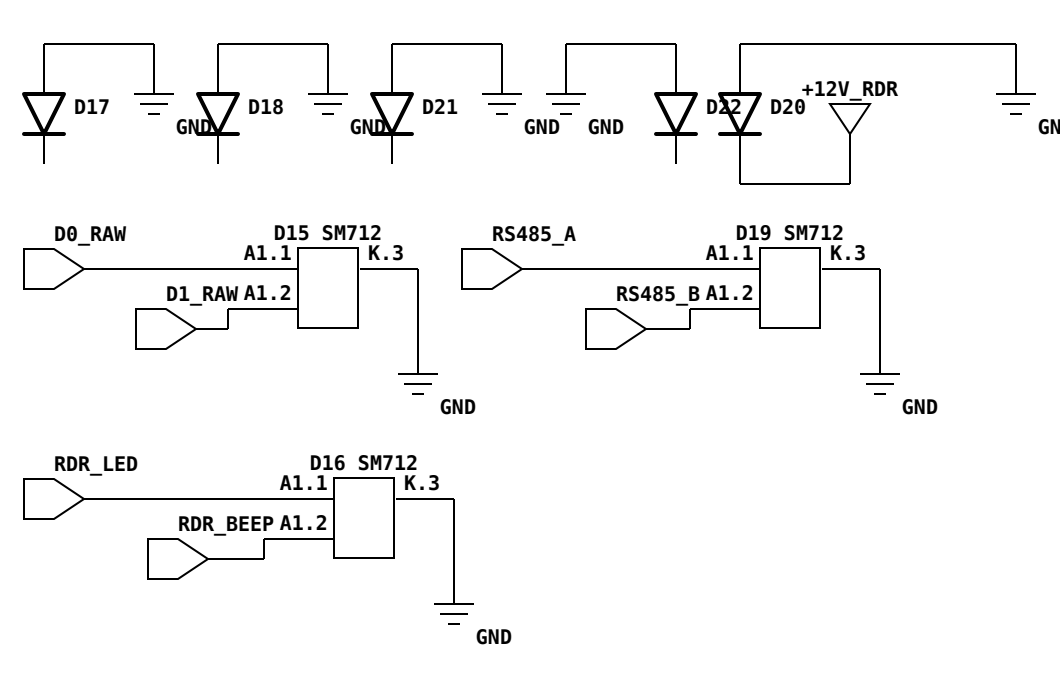

+3V3
+3V3
R37
4.7k
+3V3
R36
4.7k
I2C_SCL
J5 OLED SOCKET
1
3
2
4
I2C_SDA
GND
+3V3
J6 TFT/EPD
1
5
2
6
SPI_CS
SPI_SCK
3
7
GND
SPI_MOSI
4
8
DISP_RST
SPI_BL_BUSY
SPI_DC

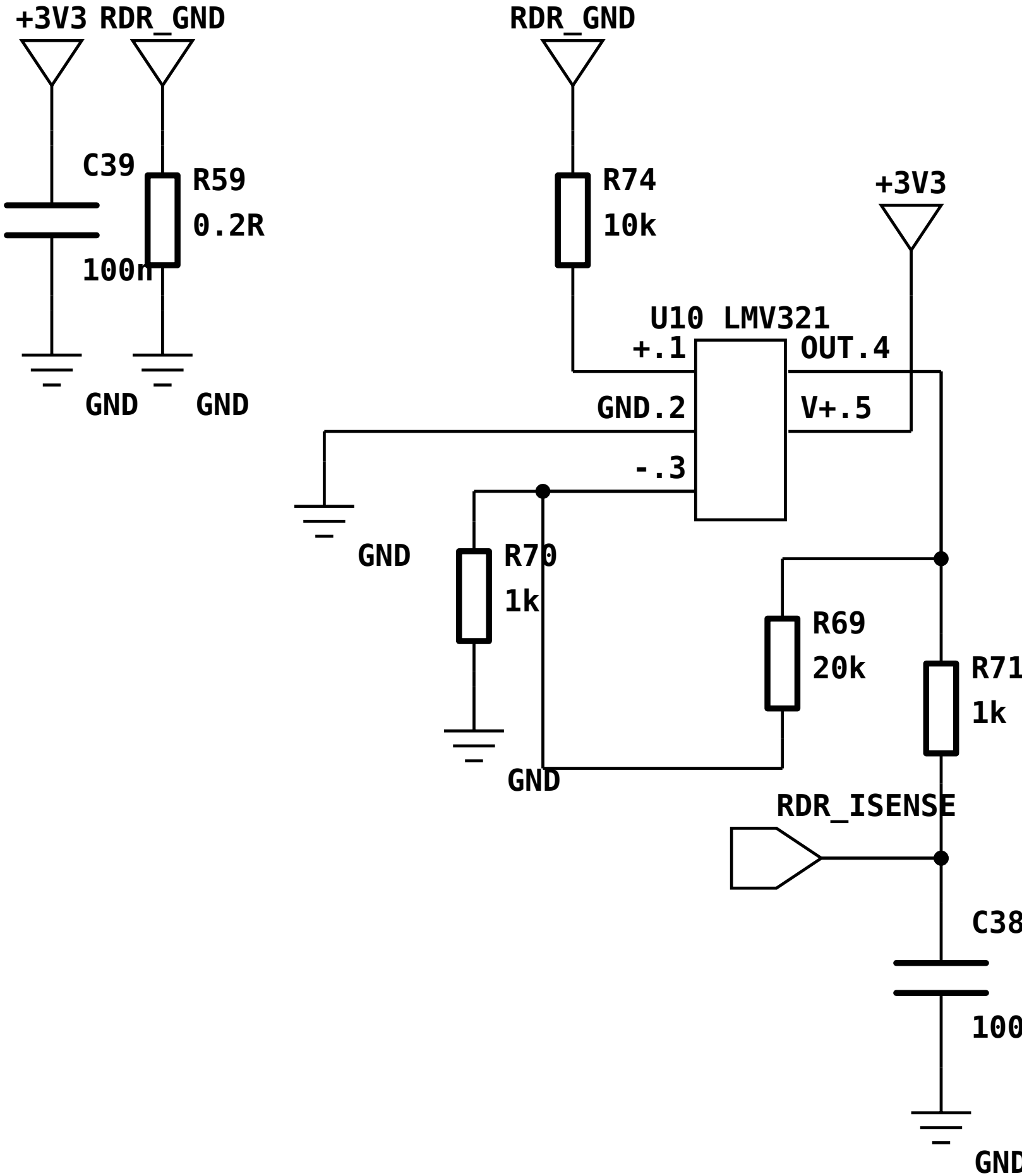

+3V3
IN4_RAW
J13 IN4 GND
1
2
R47
10k
+3V3
R43
1k
GND
Q10 2N7002
G.1
D.3
IN4
S.2
C30
10n
GND
+3V3
R46
10k
+3V3
J12 GND IN3
1
2
IN3_RAW
GND
R42
1k
Q9 2N7002
G.1
D.3
IN3
S.2
C29
10n
GND
+3V3
IN1_RAW
+3V3
J9 IN1 IN2
1
2
IN2_RAW
R44
10k
R40
1k
R45
10k
R41
1k
+3V3
Q7 2N7002
G.1
D.3
IN1
S.2
Q8 2N7002
G.1
D.3
IN2
S.2
C27
10n
C28
10n
GND
GND

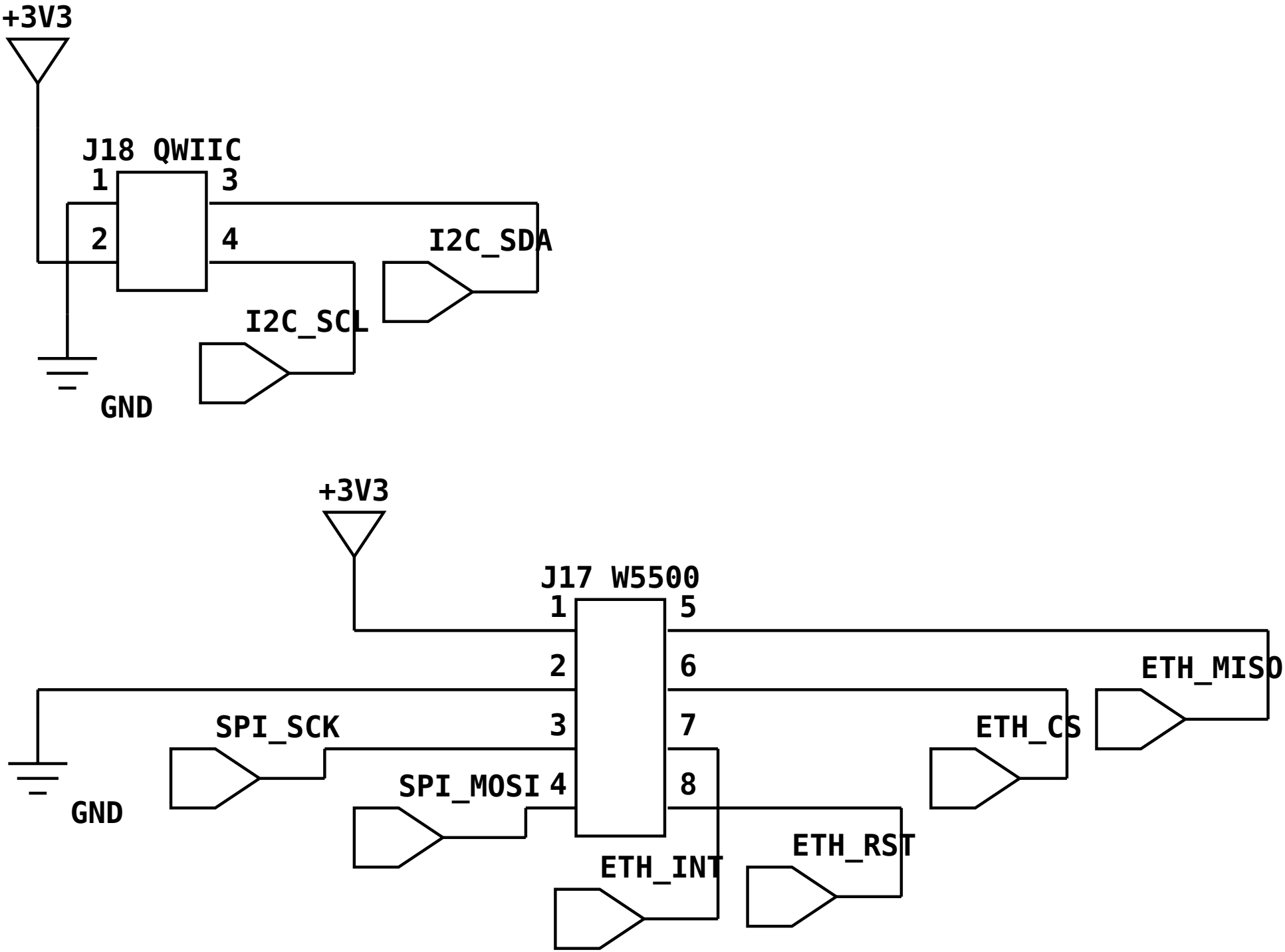

+3V3
C36
100n
GND
U8 TCA9535
~{INT}.1
A1.2
A2.3
P00.4
P01.5
P02.6
P03.7
P04.8
P05.9
P06.10
P07.11
GND.12
P10.13
P11.14
P12.15
P13.16
P14.17
P15.18
P16.19
P17.20
A0.21
SCL.22
SDA.23
VCC.24
CHRG
ETH_RST
DONE_N
EXP_INT
PGOOD5
RELAY_DRV
EXT_SENSE
R61
100k
R76
10k
R62
100k
R60
10k
RN1 4x10k
RN2 4x10k
RN3 4x10k
J19 EXPANSION
I2C_SCL
I2C_SDA


+5V
RELAY_DRV
K1 SRD-05VDC-SL-C
1
4
2
3
5J16 RELAY COM
2
J15 RELAY NC NO
1
2
R63
1k
Q12 AO3400A
G.1
D.3
S.2
R64
100k
+5V
D14
GND
GN


+3V3
+12V_RDR
RDR_GND
C37
100n
J21
RS485 12V GND
1
2
GND
+3V3
U9 SP3485
RO.1
~{RE}.2
DE.3
DI.4
GND.5
A.6
B.7
VCC.8
RS485_DE
RS485_RO
RS485_DI
R65
10k
R75
10k
+3V3
GND
RS485_B
GND
R72
390
+3V3
JP3 BIAS
1
2
R73
390
R67
20k
JP4 BIAS
1
2
GND
GND
RS485_A
J20 RS485 A B
1
2
R68
120
R66
20k
+3V3
JP2 TERM
1
2


D17
GND
D18
GND
D21
GND
GND
D22
D20
+12V_RDR
GN
D0_RAW
D15 SM712
A1.1
K.3
D1_RAW
A1.2
GND
RS485_A
D19 SM712
A1.1
K.3
RS485_B
A1.2
GND
RDR_LED
D16 SM712
A1.1
K.3
RDR_BEEP
A1.2
GND


+3V3
RDR_GND
C39
100n
R59
0.2R
GND
GND
RDR_GND
R74
10k
+3V3
U10 LMV321
+.1
OUT.4
GND.2
V+.5
-.3
GND
R70
1k
R69
20k
R71
1k
GND
RDR_ISENSE
C38
100
GND


+3V3
J18 QWIIC
1
3
2
4
I2C_SDA
I2C_SCL
GND
+3V3
J17 W5500
1
5
2
6
ETH_MISO
SPI_SCK
3
7
ETH_CS
GND
SPI_MOSI
4
8
ETH_RST
ETH_INT